

LUCAS PINEDE

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EDUCATION

CHIMIE PARISTECH-PSL, ENSCP, Paris 09/22-08/26 (expected)

Engineering degree, Master of Engineering Student

- Top graduate engineering school specialized in chemistry.
- Relevant topics: Theoretical chemistry – Statistical Thermodynamics
- Theory and experimental: Spectroscopy, Materials chemistry, Analytical chemistry
- Exchange course: Deep Learning, at [Telecom Paris](#)

PSL UNIVERSITY, Paris 09/25-08/26 (expected)

Master of Chemistry, Double diploma Master of Science student (M2)

- Relevant coursework: Data Science and AI for Chemistry – Modeling Materials – Theory of chemical reactivity – Advanced Statistical Physics (at ENS-PSL)
- Master thesis at Entalpic (AI-for-materials startup)

ESPCI PARIS-PSL, Paris 01/24-04/24

Exchange Semester Master student (M1)

- Statistical Learning – Deep Learning – Soft Matter – Biophysics – Molecular Biotechnology

AUX LAZARISTES, Lyon 09/20-08/22

Classe Préparatoire – Highly selective intensive degree in Mathematics and Physics

- Specialization: Physics and Chemistry (PC)

RESEARCH EXPERIENCE

ENTALPIC, Paris 09/25-08/26

Junior ML and Chemistry Scientist

- Building a ML pipeline for the screening of cathode materials based on voltage and conductivity properties
- Working on ML property prediction methods for chemical systems

MIT, Materials Science and Engineering, Cambridge, MA, USA 04/25-08/25

Visiting Graduate Student in Prof. R. Gomez-Bombarelli's group

- Exploited foundational force fields models with representation alignment and fine tuning for efficient training of diffusion Boltzmann Emulators
- Designed a novel inference-time approach for the direct sampling of transition state structures from diffusion models

Ecole Normale Supérieure – PSL, Paris 11/24-03/25

Research Intern in Prof. D. Laage's group

- Leveraged higher-order message-passing equivariant neural networks, with a gaussian process framework for data efficient learning of force fields
- Studied Criegee's intermediate reaction at the air-water interface using the trained equivariant MLIP: energy surface exploration (OPES, umbrella sampling)

LIMMS, University of Tokyo, Tokyo 04/24-10/24

Research Intern in Prof. S.H. Kim laboratory with Dr. A. Genot

- Developed and optimized a cancer cell capture and culture hydrogel system fully based on single stranded DNA as polymer scaffold: rheophysical tuning of the gel by designing the nanostructure – cell biophysics constraints – stability to nuclease digestion – cell culture conditions.

INSTITUTE OF CHEMISTRY FOR LIFE AND HEALTH SCIENCES, Paris 07/23

Research intern in Prof. A. Varenne's group

State of the art and optimization for the development of a laboratory on a proteomic chip

- Designed and developed an origami microfluidic paper-based analytical device (μ PADS) for protein isoelectric focusing

PUBLICATIONS

- L. Hyukjun+, S. Yang+, **L. Pinède+**, M. Steiner, Y. Du, R. Gómez-Bombarelli, A Priori Sampling of Transition States with Guided Diffusion. *39th Conference on Neural Information Processing Systems (NeurIPS 2025) Workshop on AI for Accelerated Materials Discovery, (Spotlight), +Equal contributions (co-first)*.
- **L. Pinède+**, S. Yang+, J. Nam, R. Gómez-Bombarelli, Unifying Force Prediction and Molecular Conformation Generation Through Representation Alignment. *Proceedings of the Workshop on Generative AI for Biology at the 42nd International Conference on Machine Learning, +Equal contributions (co-first)*.
- A. Cochard, **L. Pinède**, S.H. Kim, A. Genot, Controllable Reconfiguration of DNA Nanostar Hydrogel Networks via Sticky-End Ligation, *Applied Materials Today* (2026).

LEADERSHIP EXPERIENCE

CHIMIE PERSPECTIVES, Paris

02/23-01/24

Junior-Entreprise (JE) of Chimie Paristech-PSL. A R&D focused student-led, non-profit consulting firm that allows students to apply their theoretical skills by performing bibliographic or experimental studies for professional clients in the chemistry field.

The Nationale Confederation of Junior-Entreprise is the first student movement in France with 25k student entrepreneurs.

President, head of strategy and partnerships (40h/week)

Association management:

- Managed a team of 17 people with cross-disciplinary activities (HR, R&D, marketing, quality control).
- Supervised 5 R&D projects for \$25k worth of sales: defined specifications and needs with industrial and startup clients.
- In charge of the compliance to the JE strict legal framework: fully supervised the preparation JE yearly audit by external auditors; achieved top compliance

School-level leadership and student engagement:

- Organized and fully animated a morning of presentations on Corporate Social Responsibilities: part of the 1st years curriculum for students
- Organized a Conference cycle for PSL University students and staff (3 Conferences) featuring corporate and academic partners and alumni

Nation-wide leadership:

- Led the movement of the Junior-Enterprises in chemistry (~150 Students, 7 JEs)
- Co-led the organization of the region's JE congress: animated and, presented and promoted the chemistry JE movement in front of more than 500 participants and corporate partners (EY, Alten, Saint-Gobain, Arkema, L'Oreal)